108501555 林紀諜

EE3032 Introduction to VLSI Design—Final

1. (15 points) Fig. 1 shows a domino logic circuit realizing the Boolean function $Y = AB+C$. Complete the following questions.

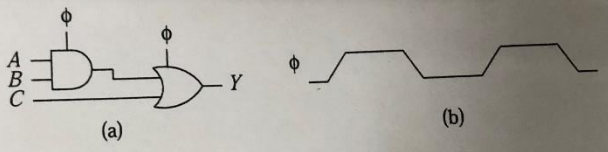


Figure 1: (a) A domino logic circuit (b) clock waveform

(a) Draw the transistor-level schematic diagram of the domino logic circuit. (5 points)

(b) Assume that the output rise time/fall time of the AND and OR gates are $3ns/2ns$ and $2.5ns/2ns$, respectively. Calculate the minimum clock period of ϕ if the duty cycle of ϕ is 50%. (5 points)

(c) Assume that each transistor has the following characteristics: $C_d = 0.1fF$; $C_g = 1fF$; $I_{Lp} = 1nA$ (PMOS leakage current); $I_{Ln} = 2nA$ (NMOS leakage current). Also, the supply voltage=1.8V and the threshold voltage of logic 1 is 1.3V. Could the circuit be operated correctly at the minimum clock period? Why? (5 points)

2. (10 points) Figure 2 shows a Schmitt trigger circuit. If you apply an input signal as shown in the left figure to the Schmitt trigger circuit, the circuit outputs the signal as shown in the right figure. Assume that $\mu_n = 2\mu_p$; the turn-on equivalent resistance of N_1 transistor is R; the simple RC delay model is used to estimate the delay and $t_r = t_f = 2RC$. Complete the following questions.

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2. (10 points) Figure 2 shows a Schmitt trigger circuit. If you apply an input signal as shown in the left figure to the Schmitt trigger circuit, the circuit outputs the signal as shown in the right figure. Assume that $\mu_n = 2\mu_p$; the turn-on equivalent resistance of N_1 transistor is R; the simple RC delay model is used to estimate the delay and $t_r = t_f = 2RC$. Complete the following questions.

(a) Let $V_{tn} = |V_{tp}| = 0.4V$, $V_{DD} = 2.0V$, $V_T^+ = 1.4V$, and $V_T^- = 0.6V$. If the size of $N_1 = 100nm/40nm$, determine the sizes of the transistors N_2, N_3, P_1, P_2, and P_3. (5 points)

(b) Fig. 2(b) shows a schmitt trigger circuit with controllable hysteresis by controlling the voltage of V_{hc}. If you want to increase V_T^+, how do you adjust the voltage of V_{hc}? (5 points)

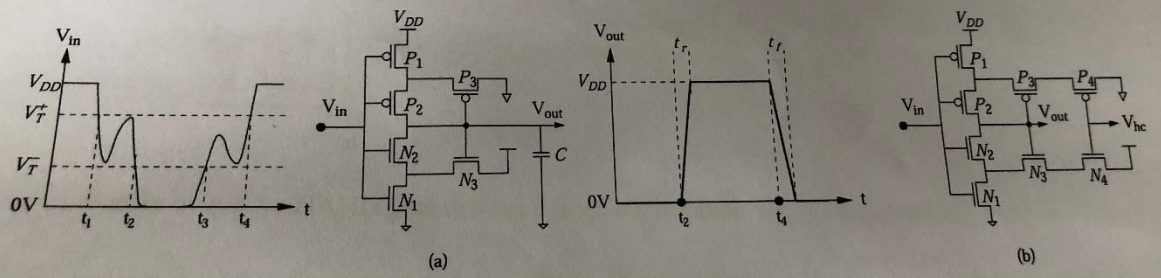


Figure 2: (a) A CMOS Schmitt trigger circuit (b) Schmitt trigger with controllable hysteresis

3. *(10 points) Draw the transistor-level schematic diagram of the Boolean function* $f = \overline{AB(C+D)}$ *using each of the following logic design styles: (1) Dynamic logic (5 points) (2) Differential split-level logic (5 points).*

4. *(10 points) Draw the transistor-level schematic diagram of the symbolic layout shown in Figure 3.*

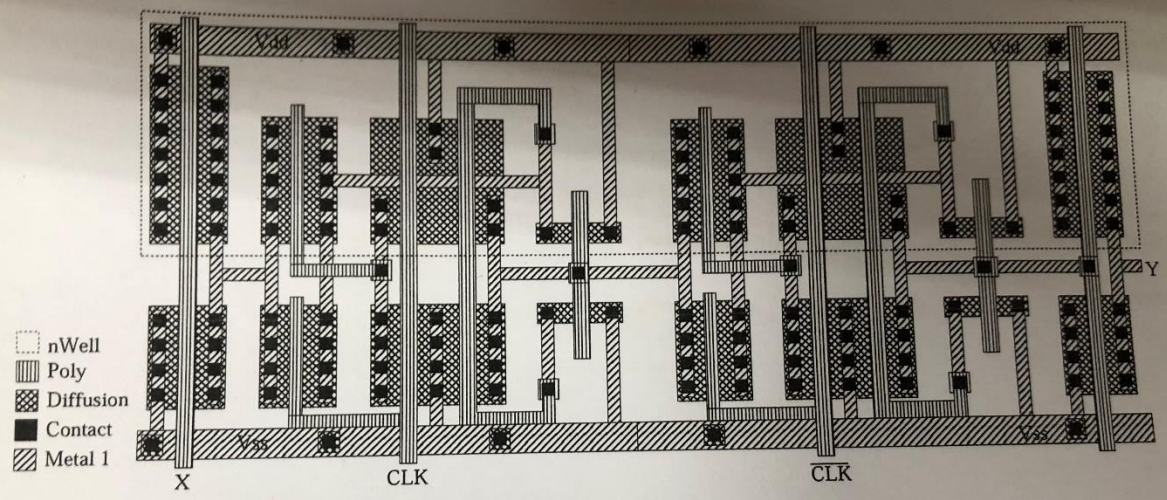


Figure 3: Symbolic layout

5. (20 points) Fig. 4(a) shows a CMOS circuit driving a large capacitance $C_L = 2.98pF$. For the us process, the input capacitance C_g of a minimum-size inverter (i.e., the size of NMOS and PM is W_{min}/L_{min}) is $1fF$. Assume that drain and wire parasitic capacitance can be neglected. A $e = 2.7$, $e^4 = 55$, $e^6 = 403$, and $e^8 = 2980$. Complete the following questions:

   (a) Design a super buffer as shown in Fig. 4(b) to drive the large capacitance such that the o load of the NAND gate A is C_g and the delay T is minimized. (7 points)

   $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$ to the circuit in terms

Metal 1
X
CLK
$\overline{CLK}$
Figure 3: Symbolic layout

5. (20 points) Fig. 4(a) shows a CMOS circuit driving a large capacitance $C_L = 2.98pF$. For the used process, the input capacitance C_g of a minimum-size inverter (i.e., the size of NMOS and PMOS is W_{min}/L_{min}) is $1fF$. Assume that drain and wire parasitic capacitance can be neglected. Also, $e = 2.7$, $e^4 = 55$, $e^6 = 403$, and $e^8 = 2980$. Complete the following questions:

   (a) Design a super buffer as shown in Fig. 4(b) to drive the large capacitance such that the output load of the NAND gate A is C_g and the delay T is minimized. (7 points)

   (b) Assume that we apply the input sequence $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$ to the circuit in terms of the opereation frequence f and the supply voltage is V_{dd}.

      i. Caculate the switching activity of the node u, v, w, and o. (5 points)

      ii. Caculate the dynamic power consumed by the super buffer. (5 points)

      iii. If the two NAND gate has the same delay, does the circuit have spurious transitions (glitches)? Why?(3 points)

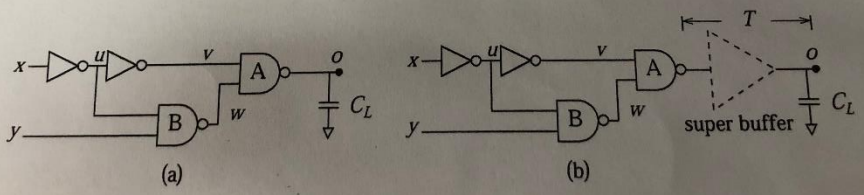


Figure 4: (a) An NAND gate driving a large capacitance. (b) A super buffer is inserted.

6. *(15 points) Figure 5 shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the*

(c) Inserting registers into the circuit such that the modified circuit can be operated at 2 times of the original operation frequency. (5 points)

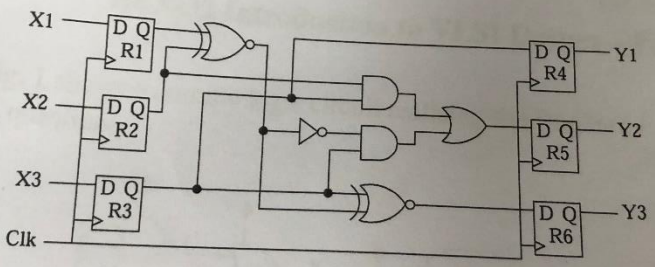


Figure 5: A sequential circuit

7. (25 points) Answer the following questions.

(a) List two requirements for drawing the layout of a cell in the standard cell library. (4 points)

(b) List two limitations of the dynamic logic. (4 points)

(c) What is the goal of the design rule? (3 points)

(d) A pseudo-NMOS logic is ...

問答題剩餘部分:

(d)A pseudo-NMOS logic is ratioed-logic. Why?

(e)What's the function of filler cell、tap cell、endcap cell、decap cell?

(f)List two technology that can shorten manufactured time and hence shorten the time to market.(應該是指 gate-array physical design 跟 sea-of-gate physical design)